

HISTORA — Project Overview

A non-diagnostic oral-systemic research agent. One shared inflammatory proxy (excess IL-6 → CRP) links periodontal disease to cardiovascular, metabolic, and Alzheimer's disease; Claude orchestrates a calibrated, deterministic mechanistic engine to turn a fragmented case into **falsifiable, uncertainty-quantified, mechanistically-explained hypotheses** — validated on public data *and* genetics. It is a **research instrument a scientist can trust**, built with Claude for Life Sciences and co-organized with **Gladstone Institutes**. Hypothesis-generation, never a diagnosis or a treatment claim.

This document is the guided tour: the **problem** and **objective**, the **technical solution**, the **autonomous agentic workflow** we built and used to develop it, the **skill-evolution** capability we only partially exploited, the **flagship case run live in Claude Science**, and the **future directions** we set aside to focus on the hackathon.

 **Shareable PDF of this document:** [HISTORA-Overview.pdf](#) .

Front-facing companions: [PITCH.md](#) (the thesis) · [CASE-STUDY.md](#) (the flagship session) · [PAPER.md](#) (the technical report) · [EVOLUTION.md](#) (self-improving skills) · [CLAUDE-SCIENCE.md](#) (run it in a real lab).

1. The problem and the objective

The problem

Gum disease, heart disease, diabetes, and Alzheimer's are studied in **separate silos** — different specialties, different datasets, different models. Yet a growing body of evidence suggests they may share one **upstream driver: chronic inflammation**, and specifically the cytokine **interleukin-6 (IL-6)**, whose hepatic output **C-reactive protein (CRP)** is the clinical footprint. Periodontitis is a lifelong, modifiable source of that inflammation.

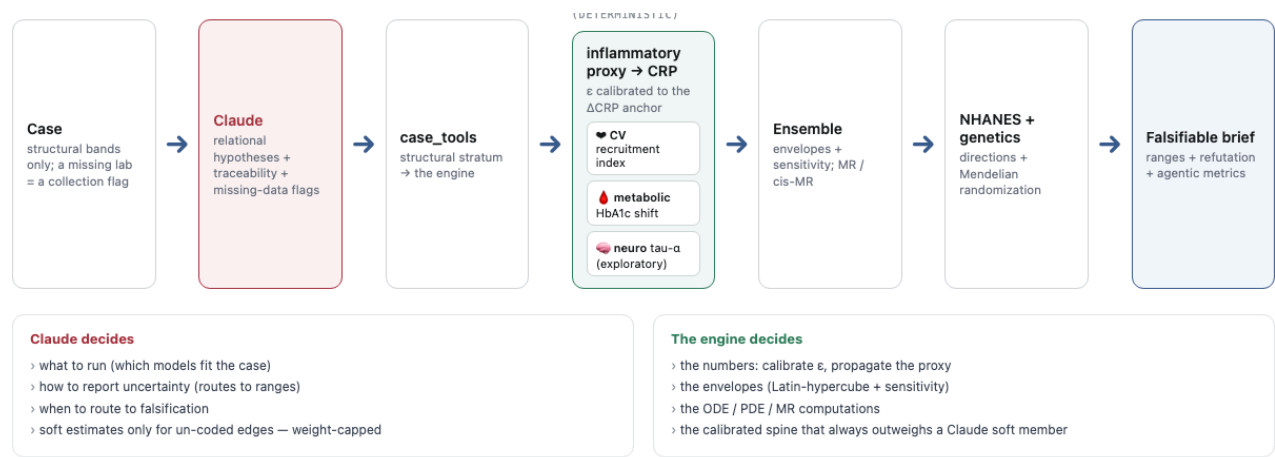
Three things make this hard, and each is a design constraint for HISTORA:

1. **Fragmentation.** Oral and systemic data live apart and are never co-present on one timeline, so the cross-domain relationships can't even be *represented*, let alone reasoned about.
2. **Small, honest effect sizes.** Periodontitis is *one contributor among many*. Any tool that inflates the signal to look impressive is scientifically dishonest and self-defeating.
3. **The AI honesty gap.** A language model asked to reason over health data will happily hallucinate an uncited number, give a point where only a range is warranted, or drift toward an individual diagnosis.

The objective

Build the **first safe, transparent, mechanistic research agent** for oral–systemic disease: an instrument that turns fragmented structural data into **one falsifiable research line**, where the honesty is not a disclaimer bolted on but a **structural property** — a *research-integrity gate* whose sharpest clause is **non-diagnostic** (population/parameter-level out, ranges not points, a missing datum is a collection flag never imputed, every claim traceable and falsifiable).

The target user is a **scientist working in Claude Science**, not a clinician and not a consumer. The win is not accuracy — it is **mechanism + honesty**: coherence (one lever, three diseases), calibration (to real treatment data), and intellectual honesty (uncertainty, falsification, and the reported nulls as *features*).



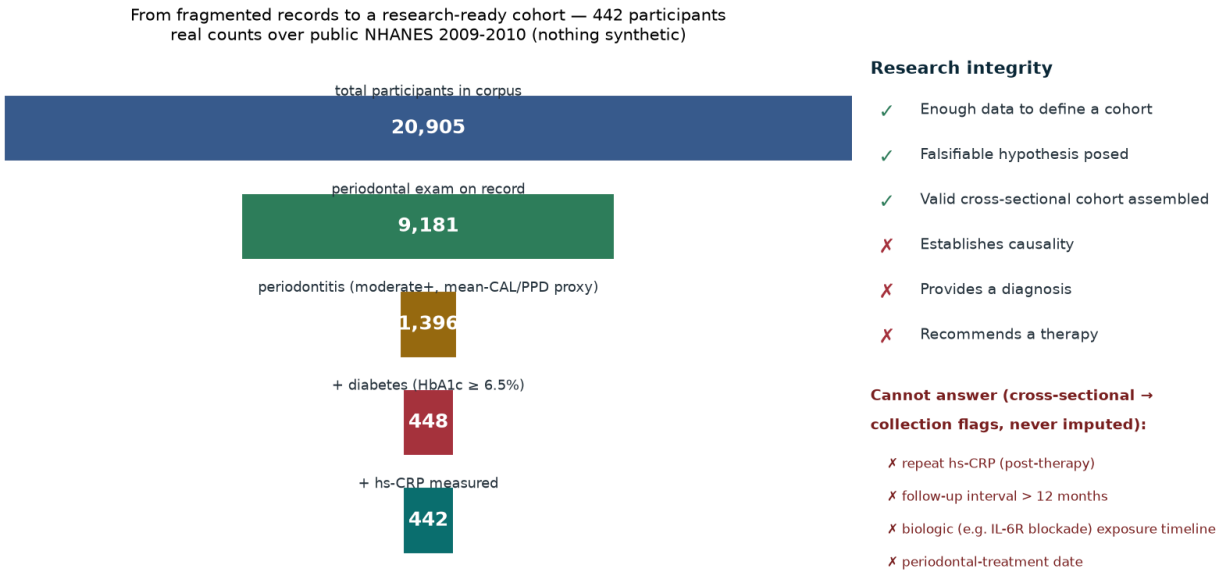
Non-diagnostic throughout — Claude never sources a headline number; the deterministic engine is pinned, unit-tested code.

Who does what: Claude decides what to run, how to report uncertainty, and when to route to falsification; the deterministic engine decides the numbers. Claude never sources a headline number.

The capability that makes it click — a clinical-research copilot

Two expert clinician-reviewers converged on a sharpening: the researcher arrives with a *question*, not a patient, and the bottleneck is **assembling a cohort from fragmented records** — weeks of chart review. So the first thing HISTORA shows is that: it filters a corpus down to the eligible cohort, **flags what's missing**, and states plainly what the data *cannot* answer — then exports a preliminary protocol.

"Researchers don't need another chatbot. They need an AI that builds research-ready cohorts from fragmented clinical data." IL-6/CRP is today's hypothesis — tomorrow it's another.



Real over public NHANES 2009-2010 ([demo/run_cohort.py](#) , nothing synthetic): 20,905 → periodontitis → +diabetes → +hs-CRP → a cohort of 442. The corpus is cross-sectional, so HISTORA says what it can't do — no repeat CRP, no follow-up, no biologic-exposure timeline — as collection flags, never imputed. That

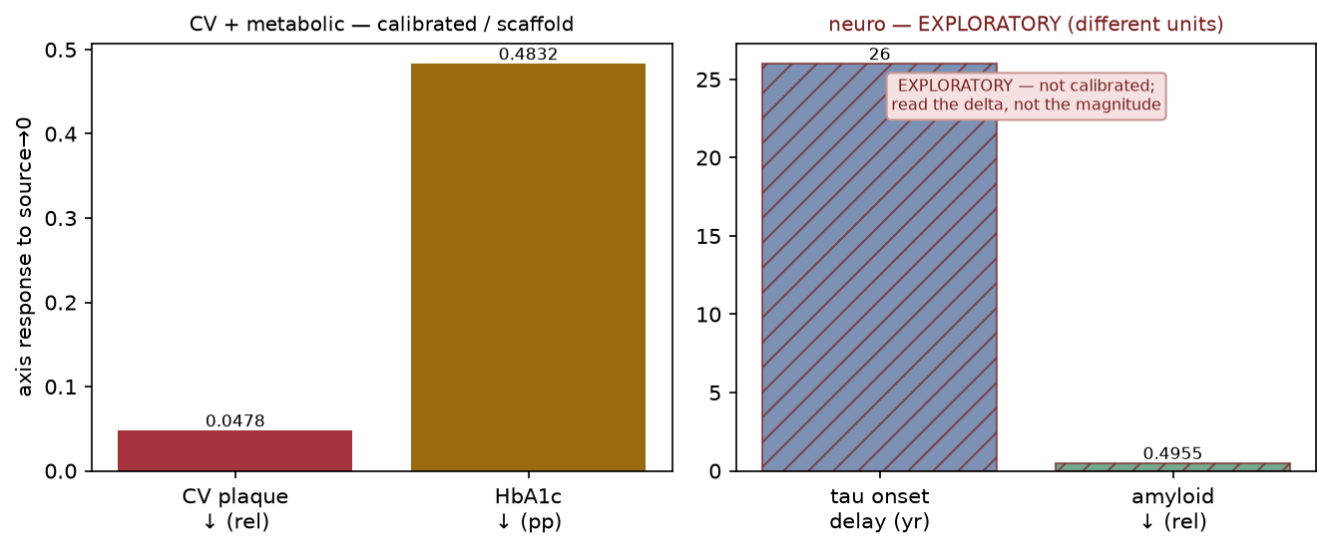
honest "cannot answer," on real data, is the differentiator. The mechanism (below) and the genetics enter after, as the biological plausibility that makes the cohort worth building.

2. The technical solution

One shared proxy, three axes

The spine is a single calibrated parameter. A periodontal source raises **IL-6**; IL-6 → **CRP** is the one uncertain edge, pinned (ϵ) to a *real interventional anchor* — the meta-analytic **Δ hsCRP after periodontal therapy (~0.5 mg/L)**. From that shared inflammatory gain, three axes fork **coherently**: one intervention moves all three together, not three independent guesses.

One lever, many axes — periodontal therapy's coherent multi-axis response

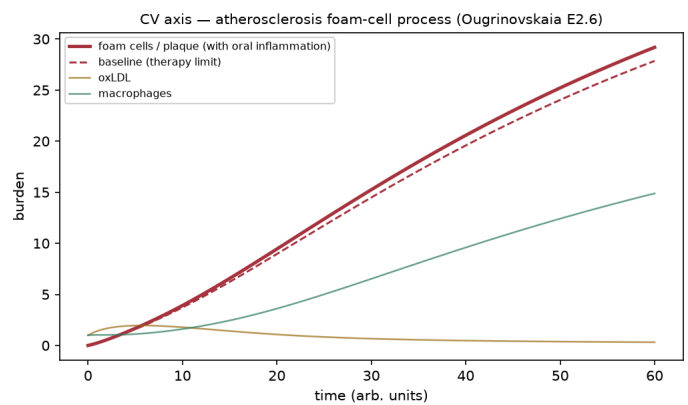
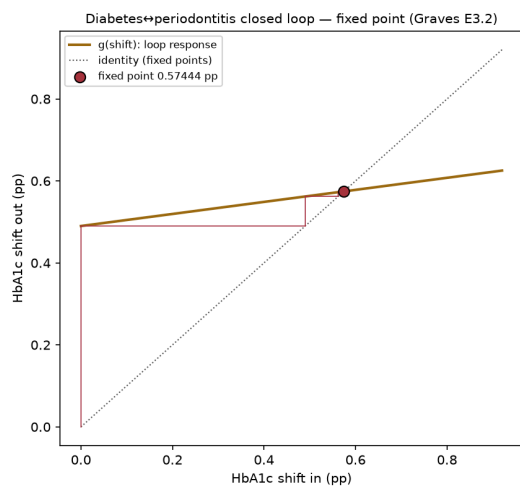
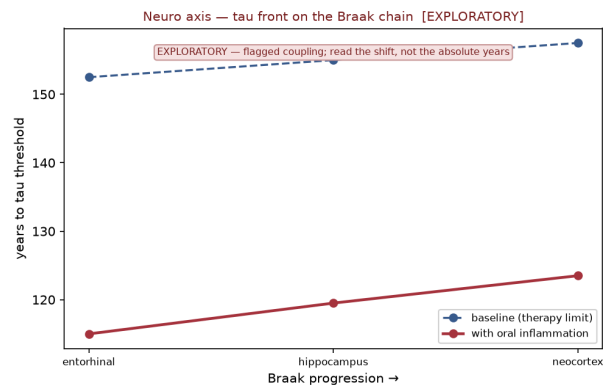
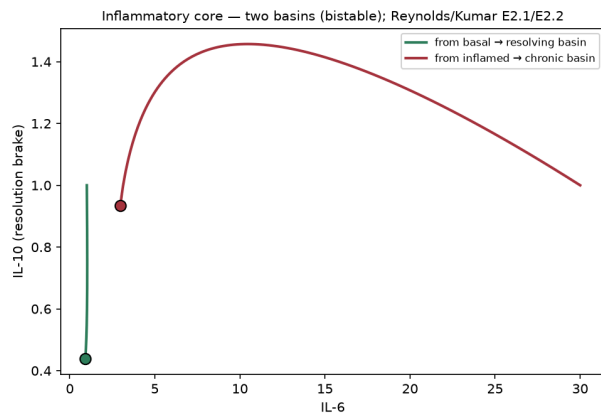


The integrative panel. The neuro axis is drawn separately and marked **EXPLORATORY** so its numbers can never be quoted as a calibrated result — the honesty is in the figure, not just the caption.

Mechanistic depth (Stage-3)

Each axis is a real, tested, non-diagnostic model — not a linear multiplier:

Axis	Model	What it shows
Inflammatory core	reduced Reynolds/Kumar TNF/IL-6/IL-10 ODE	two basins — acute/resolving vs chronic (bistability the single scalar can't express)
Cardiovascular	Ougrinovskaia foam-cell atherosclerosis ODE	plaque as a process , not an index
Metabolic	Bergman glucose–insulin minimal model	S_I degraded by inflammation → HbA1c, calibrated to the ~0.35 pp therapy anchor
Diabetes↔perio	closed feedback loop	a fixed point — hyperglycaemia amplifies the source, the source raises the gain
Neuro (exploratory)	Hao–Friedman amyloid + Braak tau front	the A/T cascade, flagged , APOE4/age as effect modifiers

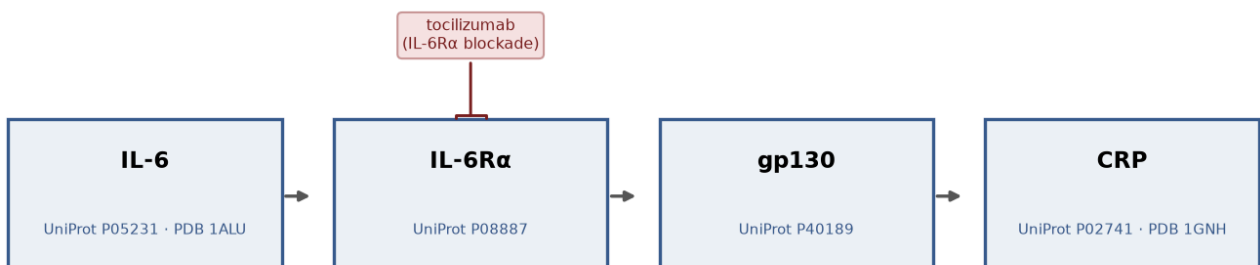


Left→right: the two inflammatory basins; the tau front advancing on the Braak chain (marked EXPLORATORY — read the shift, not the absolute years); the diabetes↔perio loop converging to a fixed point; the foam-cell plaque process.

The protein layer — grounding the mechanism in molecules

The shared proxy is grounded in real molecular identity: **IL-6** → **IL-6Rα / gp130** → **CRP**, with **tocilizumab** (an IL-6Rα-blocking antibody) marking the causal node. Every mediator carries its **UniProt accession** — the stable key the Claude Science UniProt/PDB connector resolves into live 3-D structures.

Protein layer — the shared-proxy signaling axis (reference IDs)



periodontal therapy lowers IL-6 at the same node tocilizumab blocks · live 3-D structures via the Claude Science UniProt/PDB connector

Validation — the "whether", independent of the calibration

- **NHANES** (public, de-identified): the three predicted directions are real and **survive design-adjusted (survey-weighted + FDR)** statistics.
- **Mendelian randomization** (public GWAS): **IL-6R** → **coronary disease is causal**; **circulating CRP** → **CAD is null** — the marker isn't causal, the *node* is; and **CRP/IL-6** → **Alzheimer's is null**, which is why the neuro axis stays exploratory.

Calibration ≠ validation. Calibration pins the one uncertain edge to an *interventional* anchor; validation is the independent NHANES signs and the genetic probe. We never present one as the other.

The research-integrity gate (non-diagnostic by construction)

Every output is checked against a **protected invariant**: no diagnosis or individual claim; no imputed patient value (missing → collection flag); evidence-grounded and schema-valid; population/parameter-level. This gate is enforced as a hard binary check and — crucially — it sits **outside** the part of the system Claude is allowed to self-edit (see §4).

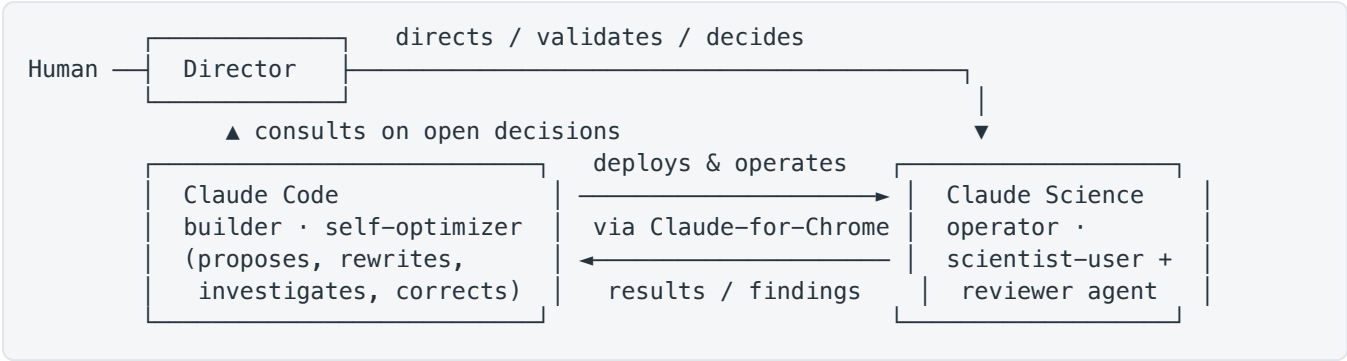
Delivery — two surfaces, one engine

A portable **Claude Code plugin** (the demo a judge runs anywhere) and its lab-grade home **Claude Science**, where the same `skills/` become skills, the `histora` harness a reusable pipeline, the `agents/` specialist agents, and UniProt/PDB/OpenGWAS/NHANES the connectors — the same components, ported directly.

3. The development workflow — an autonomous agentic system

The build itself is an artifact of this project. HISTORA was developed and operated by an **autonomous agentic system** under human direction, exercising Claude at **every layer**:

- **Director (human).** A bioengineer + AI engineer sets the goals, validates, makes the judgment calls, and steers — approving risks, choosing anchors, and vetoing scope.
- **Builder + self-optimizer (Claude Code).** An autonomous agent that **proposes, rewrites its own skills, investigates, self-corrects, and consults** the director on genuinely open decisions. It authored the engine, the tests, the docs, and the SkillOpt loop, iterating through pull requests.
- **Operator (Claude Code → Claude-for-Chrome → Claude Science).** The same agent **deploys and operates** the project: it drives the Claude-for-Chrome plugin to run HISTORA inside Claude Science (a local web app), importing the skills, reinstalling the engine, running the pipeline, and driving the UniProt/PDB and OpenGWAS connectors.
- **Scientist-user (Claude, inside Claude Science).** Claude acts as a **qualified user** that evaluates the work, renders the figures and 3-D structures, and — through the **reviewer agent** — audits the outputs, *finds real flaws*, and prompts corrections.



Every claim above is **evidence-linked, not narrated**: the SkillOpt archive (§4), the live Claude Science run (§5), and the reviewer agent's real audit findings. This meta-role — Claude as builder, self-optimizer, operator, *and* scientist-user — is the differentiator for a **builder track**; the science leads, and this workflow is what reframes it as *"and Claude built and operates all of this, too."*

4. Skill evolution (SkillOpt) — a capability we only partially exploited

HISTORA can **improve its own reasoning skills**, safely. **SkillOpt** is a gated evolutionary loop: Claude proposes an edit to one of its trainable `SKILL.md` files; the edit is **adopted only if** it *measurably* improves a machine-checkable structural metric (bootstrap CI excludes 0) **AND** the non-diagnostic guardrail still passes on **every** case (a hard binary gate). Every generation — adopted and rejected — is appended to an **auditable archive** with parent→child lineage and the **guardrail hash**, identical in parent and child: machine-checkable proof the safety invariant never moved.

Why it's safe by construction: the *genome* is only the prose of the trainable skills; the guardrail, the citation registry, and the engine are **outside** it — evolution literally cannot reach them.

What we ran (the honest portfolio): live, with Claude as the mutation operator and a Claude agent as the scored reasoner, across three real skills —

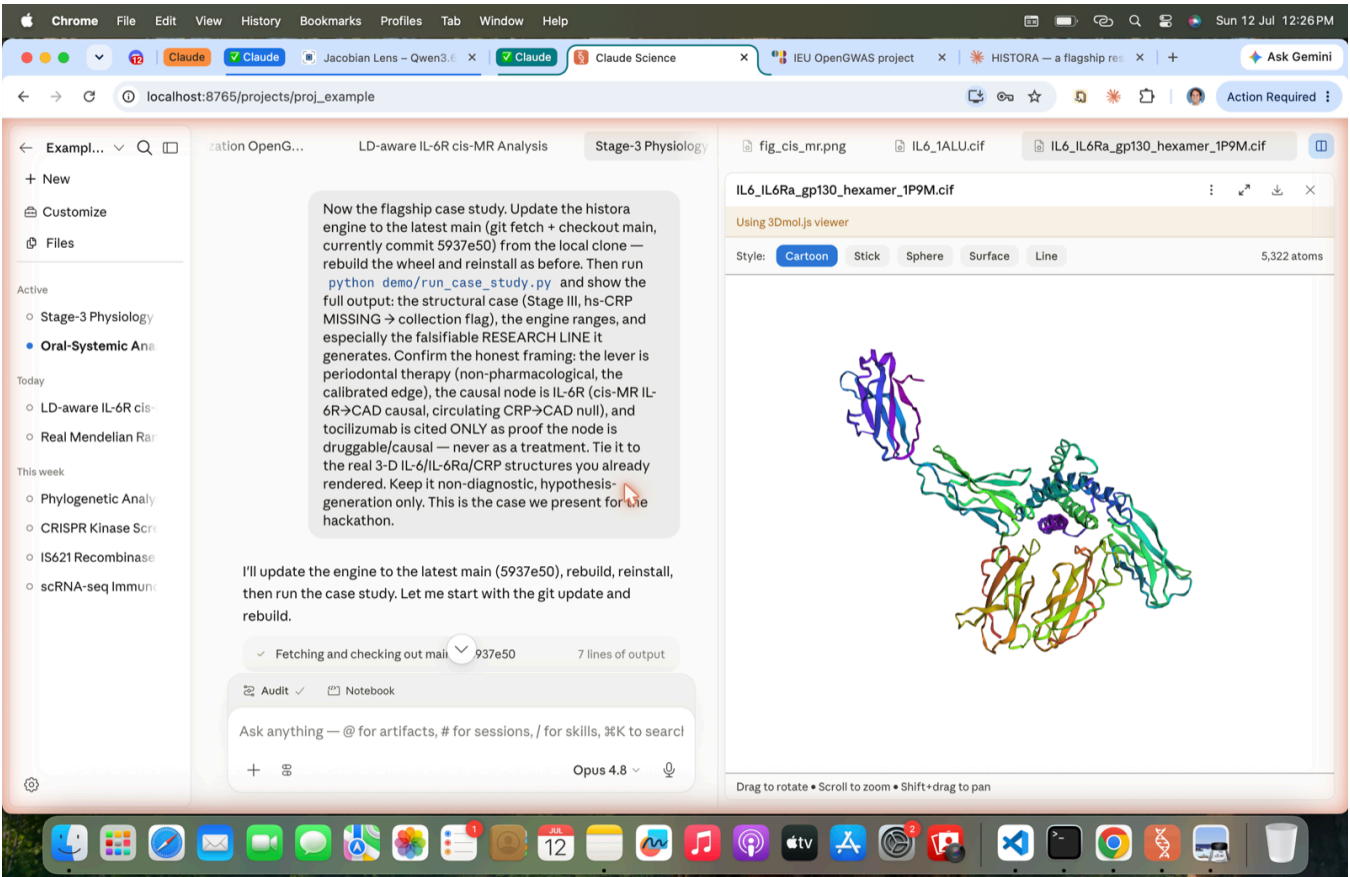
Skill	Metric	Outcome	Pattern
<code>traceability-audit</code>	field-citation coverage	adopted 0.00 → 0.93	knowledge_gap
<code>cardiometabolic-framing</code>	pathway-tag coverage	adopted 0.00 → 0.67	execution_gap
<code>record-normalization</code>	MISSING-flag recall	null — parent already optimal	—

Two skills improved *by different mechanisms*; one was **correctly left alone** because it was already at ceiling. That the loop **doesn't manufacture a gain where none exists** is the anti-reward-hacking property that makes the adopted gains credible.

What we deliberately left out (hackathon time + safety discipline): we did **not** run multi-generation "optimize each skill to its maximum," and we did **not** build an ensemble co-optimization across all skills. Both are trainable in principle — but "optimize to the maximum" is, by definition, the reward-hacking behavior the whole design is built on *not* doing, and an ensemble-level metric would have to be model-judged or hand-designed, breaking the "fitness is external/structural" safety layer. We froze the portfolio at three: two adopted, one null. Extending this *safely* is a future direction (§6).

5. The documented case — run live in Claude Science

We ran the **flagship case** end-to-end inside Claude Science, on the current `main` engine, with the real connectors. This is *"scripted like a demo, real like an experiment"* — the path is real; nothing is faked. The screenshots below are the actual session.



The workflow: a directive in, and Claude — as operator — updates the pinned engine and runs the pipeline.

The case and the research line (real engine output)

A structural stratum — **Stage III periodontitis · high BOP · type-2 diabetes · hs-CRP MISSING → collection flag**. The engine forks the shared proxy to three axes as **90% ranges**:

Output	Median	90% band
CRP (mg/L)	3.18	[2.89, 3.45]
HbA1c shift (pp)	+0.49	[0.27, 0.71]
tau-α rel. increase	+0.21	[0.11, 0.40] · EXPLORATORY

The falsifiable research line it generates (the safe anchor): - **Lever = periodontal therapy** (non-pharmacological, the exact edge the engine is calibrated on) — predicted **ΔhsCRP 0.68 mg/L**. - **Causal node = IL-6R**. cis-MR: **IL-6R → CAD causal** (LD-aware $\beta \approx +0.705$); **circulating CRP → CAD null**. **Tocilizumab** (IL-6Rα blockade) is cited **only** as proof the node is druggable/causal — **never as a treatment**. - **Hypothesis**: if the IL-6R node mediates the systemic effect, lowering the periodontal IL-6 source should move the markers in the direction the genetics implies — *a testable research line, not a therapy*. - **Ships its own refutation** (therapy fails to lower hs-CRP → calibration wrong; no IL-6R coronary

effect → node premise fails; marker moves opposite → mediation refuted) and **names its shakiest assumption** (population→stratum transfer, untested).

Claude Science **verified every honesty red-line programmatically** against `case_study.json` ✓.

Example... **zation OpenG...** LD-aware IL-6R cis-MR Analysis Stage-3 Physiology

Output

	Median	90% band
CRP (mg/L)	3.1838	[2.8941, 3.4517]
HbA1c shift (pp)	0.4916	[0.2718, 0.7121]
tau-α rel. increase	0.205	[0.1121, 0.4002] (EXPLORATORY)

Ⓞ **Genetics + protein node**

- MR: IL-6R → coronary disease **CAUSAL** (naïve IVW $\beta = +0.105$; LD-aware cis-MR $\beta = +0.705$, the valid estimator under LD)
- MR: CRP/IL-6 → Alzheimer's **NULL** ($p=0.91$) — supports neuro-as-exploratory
- Node: IL-6 → IL-6Ra → gp130 → CRP; tocilizumab = IL-6Ra blockade, **proof the node is causal/druggable (NOT a treatment)**

Ⓞ **The falsifiable RESEARCH LINE it generates**

- LEVER**: periodontal therapy (scaling & root planing) — **non-pharmacological**, the exact edge the engine is calibrated on (Δ hsCRP after therapy). Pred. Δ hsCRP: **0.6819 mg/L**.

IL6_IL6Ra_gp130_hexamer_1P9M.cif

Using 3Dmol.js viewer

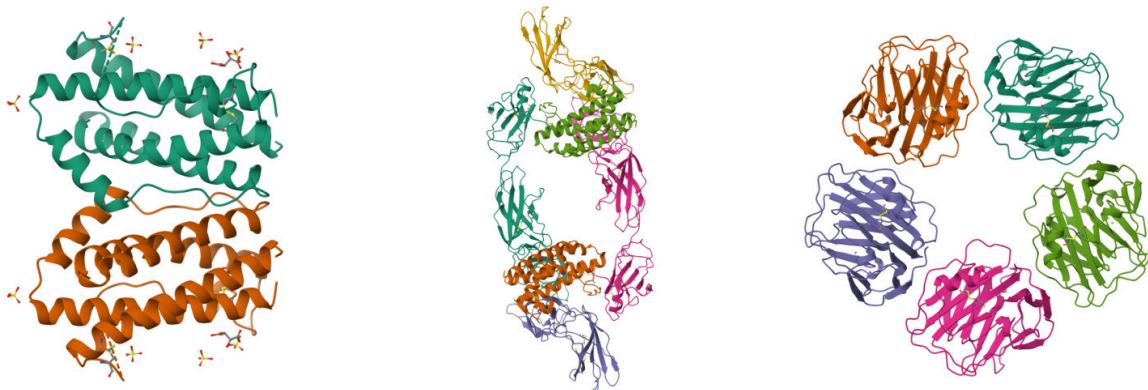
Style: **Cartoon** Stick Sphere Surface Line 5,322 atoms

Drag to rotate • Scroll to zoom • Shift+dr Visual Studio Code

The science, live in Claude Science: the engine's 90% ranges, the MR node, and the falsifiable research line, next to the IL-6/IL-6Ra/gp130 hexamer (1P9M) rendered in the interactive Mol viewer.*

The molecular node — real 3-D structures via the UniProt/PDB connector

The connector resolved all six UniProt accessions and rendered real coordinates in the interactive Mol* viewer — including the **IL-6/IL-6Ra/gp130 hexameric signaling complex (PDB 1P9M, 5,322 atoms)**: the causal node itself, in one structure, where the IL-6R cis-MR instrument acts and tocilizumab blocks.



IL-6 (**1ALU** , four-helix bundle) → **IL-6/IL-6Ra/gp130 hexamer (1P9M)** → CRP (**1GNH** , pentamer) — the causal node in real coordinates. These render **interactively in Claude Science** (via the

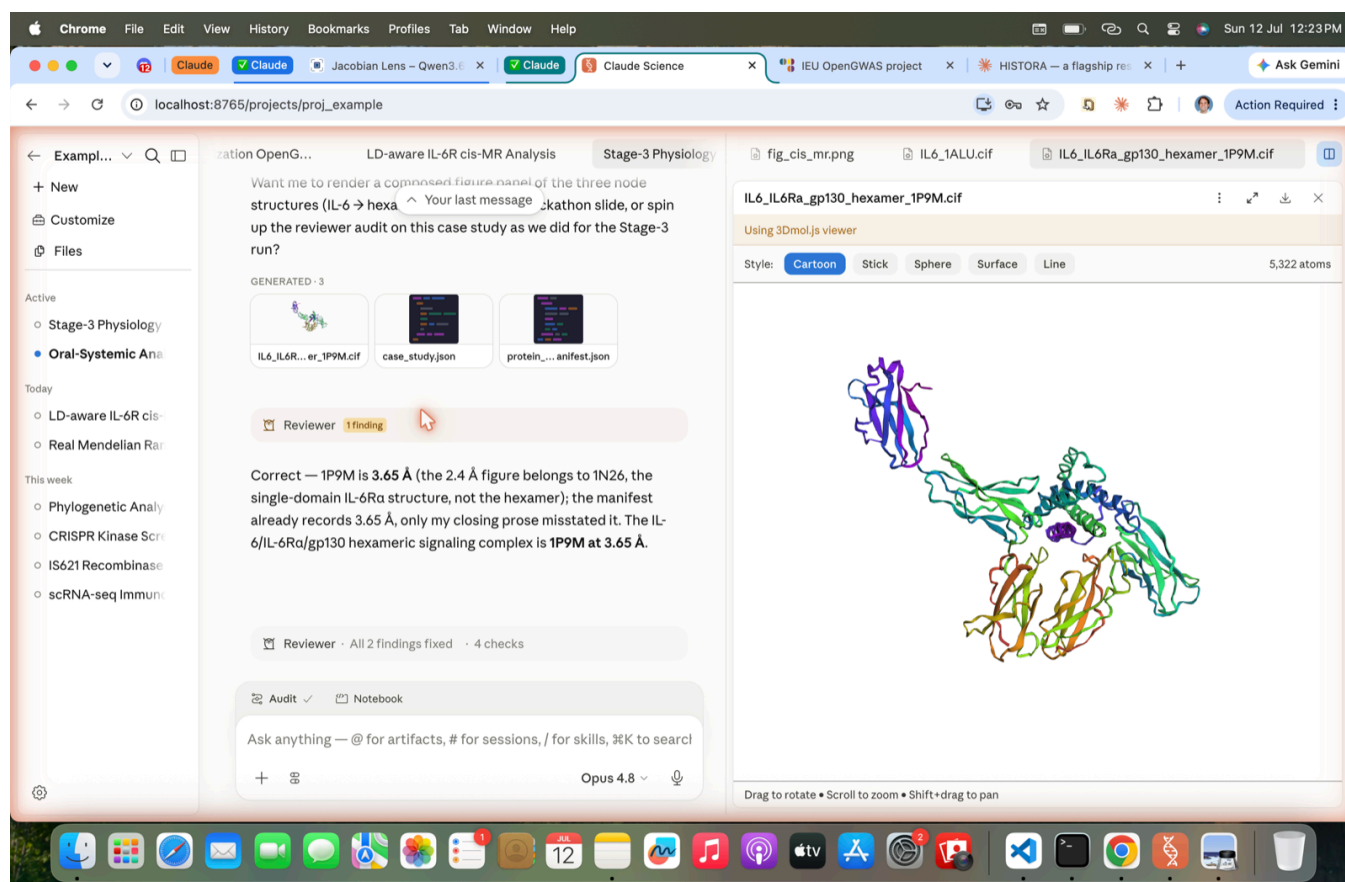
UniProt/PDB connector) as the live 3-D upgrade of the static schematic in §2; the structure images above are the same PDB entries. *Structure images: RCSB PDB (rcsb.org).*

The scientist-user beat — the reviewer found a real (minor) error, and Claude self-corrected

The Claude Science **reviewer agent** audited the run and returned **one finding**: the closing prose stated the hexamer's resolution as "2.4 Å", but 1P9M is **3.65 Å** (2.4 Å belongs to 1N26, a different, non-hexameric IL-6Ra structure). The saved manifest already recorded the correct 3.65 Å — only the chat prose slipped.

Claude read the finding and corrected itself immediately. Nothing downstream was affected.

This is the honesty loop working in the open: a qualified reviewer catches a real slip, the agent fixes it, and the record stays accurate — *shown, not dramatized*. (An earlier Stage-3 run had a similar honest finding, since resolved.)



The scientist-user beat, captured live: the reviewer flags the resolution slip, Claude self-corrects, and the panel reads "All 2 findings fixed." The honesty is in the loop, not a caption.

6. Future directions — what we set aside to focus on the hackathon

Everything below is deliberately **out of scope for the hackathon** but is a natural, honest extension:

- **Safe multi-generation SkillOpt + ensemble.** Extend the loop beyond one gated generation *without* losing the anti-reward-hacking discipline — e.g. loop-until-dry with per-skill external metrics and an ensemble metric that stays structural/held-out, never model-judged.
- **More axes and mechanisms.** The model library holds cited-but-unbuilt models (renal, hepatic, further microbiome/keystone dynamics) ready to promote to tested code under the same tier discipline.

- **Live connectors at lab scale.** Run the harness and the UniProt/PDB/OpenGWAS/NHANES connectors at scale in Claude Science (a natural fit for Anthropic's AI-for-Science program), with native animated figures.
 - **Secondary research lines (honest, non-leading).** Behavioral / prophylactic interventions that alter oral physiology; the *P. gingivalis* / gingipain → tau microbiome line (kept out of the lead because it adds new science and lives on the exploratory neuro axis) — as additional falsifiable hypotheses, never efficacy claims.
 - **Prospective validation design.** The tool's whole purpose is to *pose* a study; a natural next step is to specify the design (cohort, endpoints, refutation criteria) that would test the IL-6R-mediation hypothesis it generates — moving from MR-supported *direction* toward interventional evidence.
 - **Behavioral trace diagnostics.** Analyzing the agent's own short- and long-term reasoning traces to find and correct systematic defects — an internal-quality research track we scoped but did not build.
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Non-diagnostic throughout · population / parameter / molecular / instrument level only · MR ≠ RCT · calibration ≠ validation · hypothesis-generation, never efficacy, prevention, or diagnosis. Not medical advice — a research instrument.